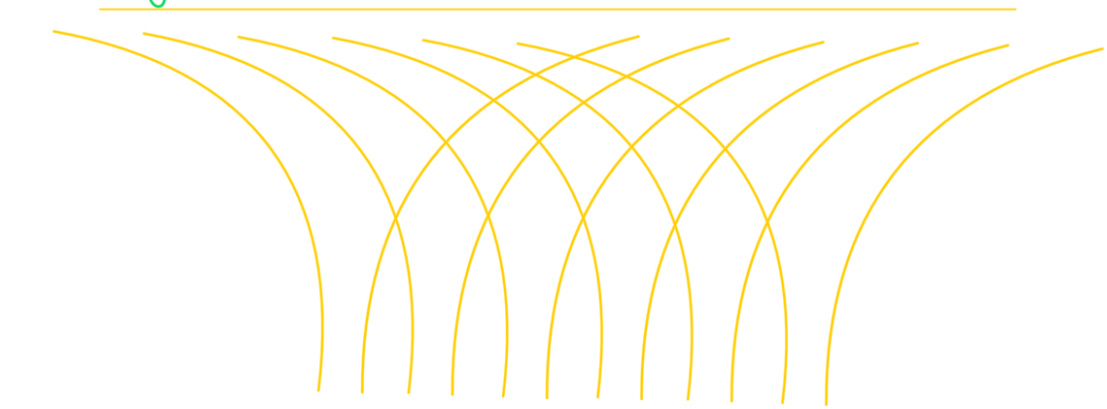
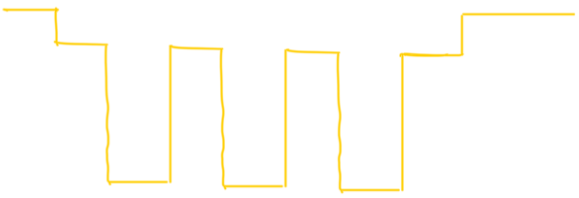
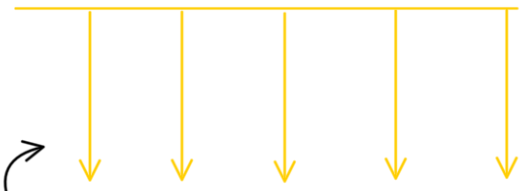
Kronig prenney model



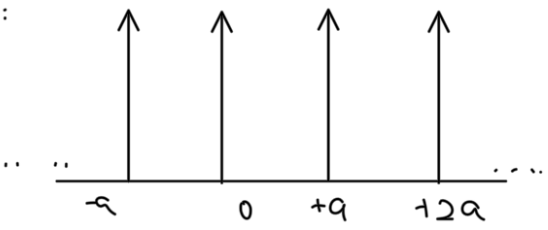


(or)



both give same, we'll do this though

for simplification, we take:



BLOCH's Theorem:

$$\psi(x) = e^{ikx} u(x) \; ; \; u(x+a) = u(x)$$

$$\Rightarrow \psi(x+a) = e^{ika} \psi(x)$$

for us $u(x) = \beta \sum_{v=-\infty}^{\infty} \delta(x - va)$

TISWE: $-\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + u(x)\psi(x) = E\psi(x)$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + \beta \sum_{v=-\infty}^{\infty} \delta(x - va) = \epsilon\psi(x)$$

we solve for $0 < x < a$

$$\Rightarrow -\frac{\hbar^2}{2m} \frac{d^2\psi(x)}{dx^2} + 0 = \epsilon\psi(x)$$

$\psi(a_-) = \psi(a_+) \rightarrow \psi$ has to be continuous

But $\psi(a_+) = e^{ika}\psi(0_+) = \psi(a_-)$

$0 < x < 2a$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \beta\delta(x-a)\phi(x) = E\phi(x)$$

interprating $a-\alpha < x < a+\alpha \quad \alpha \to 0$

$$-\frac{\hbar^2}{2m}\int_{a-\alpha}^{a+\alpha}\frac{d^2\psi}{dx^2}dx + \int_{a-\alpha}^{a+\alpha}\beta\,\delta(x-a)\psi(x)\,dx = E\int_{a-\alpha}^{a+\alpha}\psi(x)\,dx \to 0$$

$$\lim_{\alpha\to 0} \Rightarrow \frac{d\psi}{dx}(a_+) - \frac{d\psi}{dx}(a_-) = \frac{2m\beta}{\hbar^2}\psi(a)$$

$$\frac{d\psi}{dx}(a_+) = e^{ika}\frac{d\psi}{dx}(0_+) \rightarrow \text{from bloch theorm}$$

$$\Rightarrow e^{ika}\frac{d\psi(0_+)}{dx} - \frac{d\psi}{dx}(a_-) = \frac{2m\beta}{\hbar^2}e^{ika}\psi(0_+)$$

two boundary conditions in 0_+ and a_-

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi(x) \Rightarrow \psi(x) = A\cos qx + B\sin qx$$

$$q = \sqrt{\frac{2mE}{\hbar^2}}$$

Boundary:

$\rightarrow e^{ika}\psi(0_+) = \phi(a_-) \Rightarrow e^{ika}A = A\cos qa + B\sin qA$ — (1)

$\rightarrow q[e^{ika}B + A\sin qa - B\cos qa] = \frac{2m\beta}{\hbar^2}a\,e^{ika}A$

$\Rightarrow q[e^{ika}B + A\sin qa - B\cos qa] = 2\gamma e^{ika}A$ — (2)

$$\gamma = \frac{m\beta a}{\hbar^2}$$

(1) $\Rightarrow A/B = \sin qa / [e^{ika} - \cos qa]$

(2) $\Rightarrow A/B = \dfrac{e^{ika} - \cos qa}{\frac{2\gamma}{qa}e^{ika} - \sin qa}$

) equate

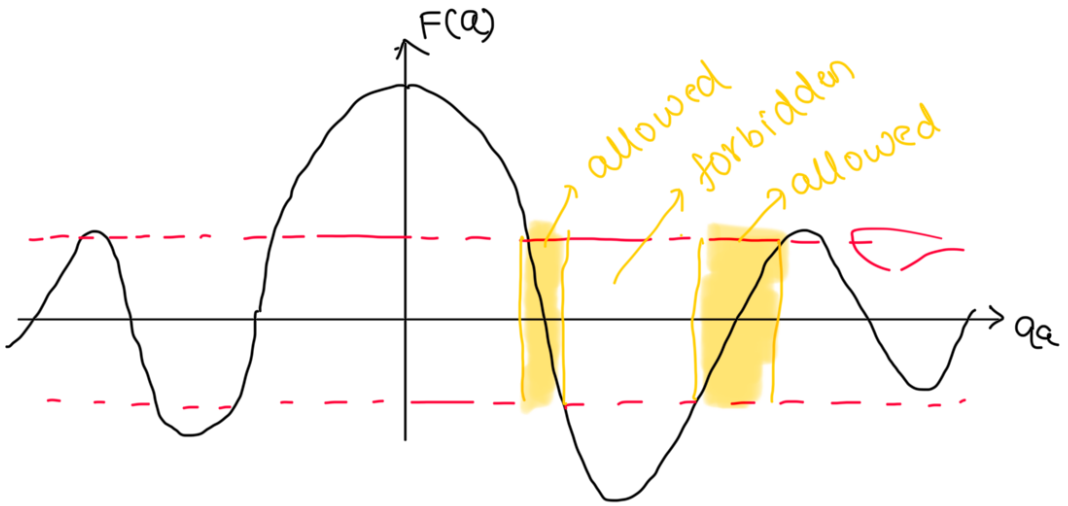
$$\Rightarrow e^{2ika} - 2e^{ika}\left[\cos qa + \beta \frac{\sin qa}{qa}\right] + 1 = 0$$

$$\Rightarrow \cos qa + \beta \frac{\sin qa}{qa} = \frac{1}{2}\left[e^{ika} + e^{-ika}\right] = \cos ka$$

$$\Rightarrow \boxed{\cos qa + \beta \frac{\sin qa}{qa} = \cos ka}$$

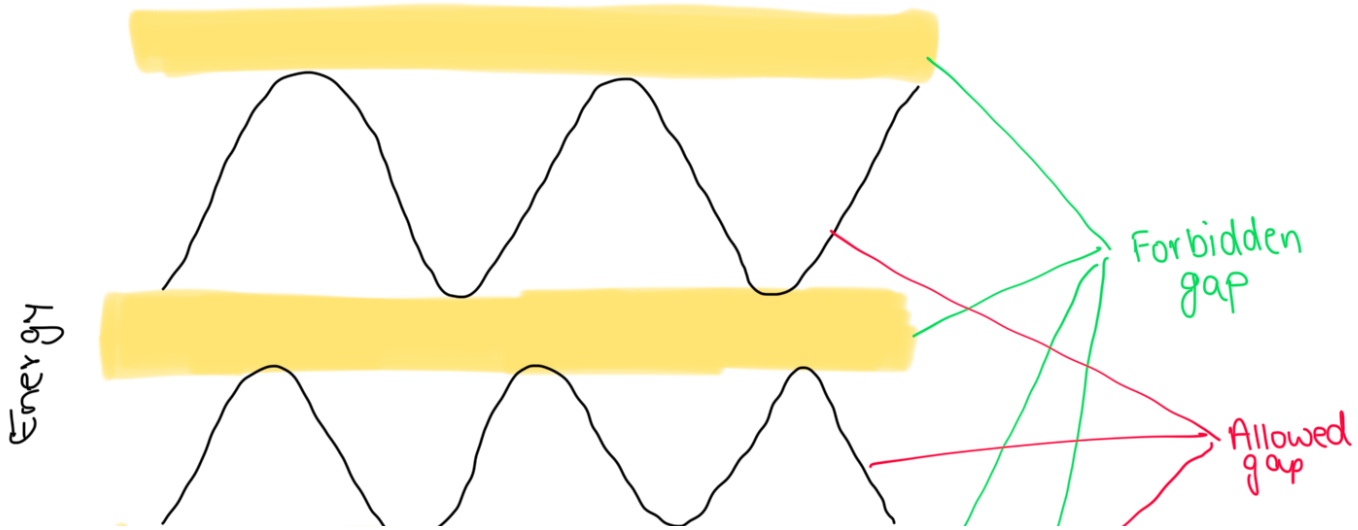
$F(q)$

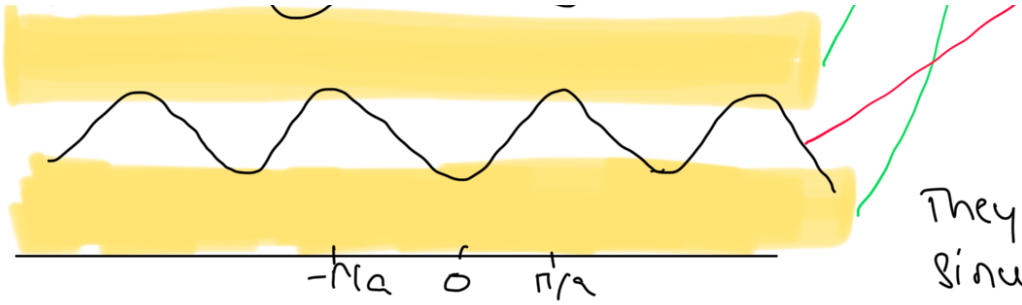
$$F(q) = \cos qa + \beta \frac{\sin qa}{qa} \qquad F(q) = \cos ka$$



Some q are allowed (means allowed energies) and rest are forbidden.

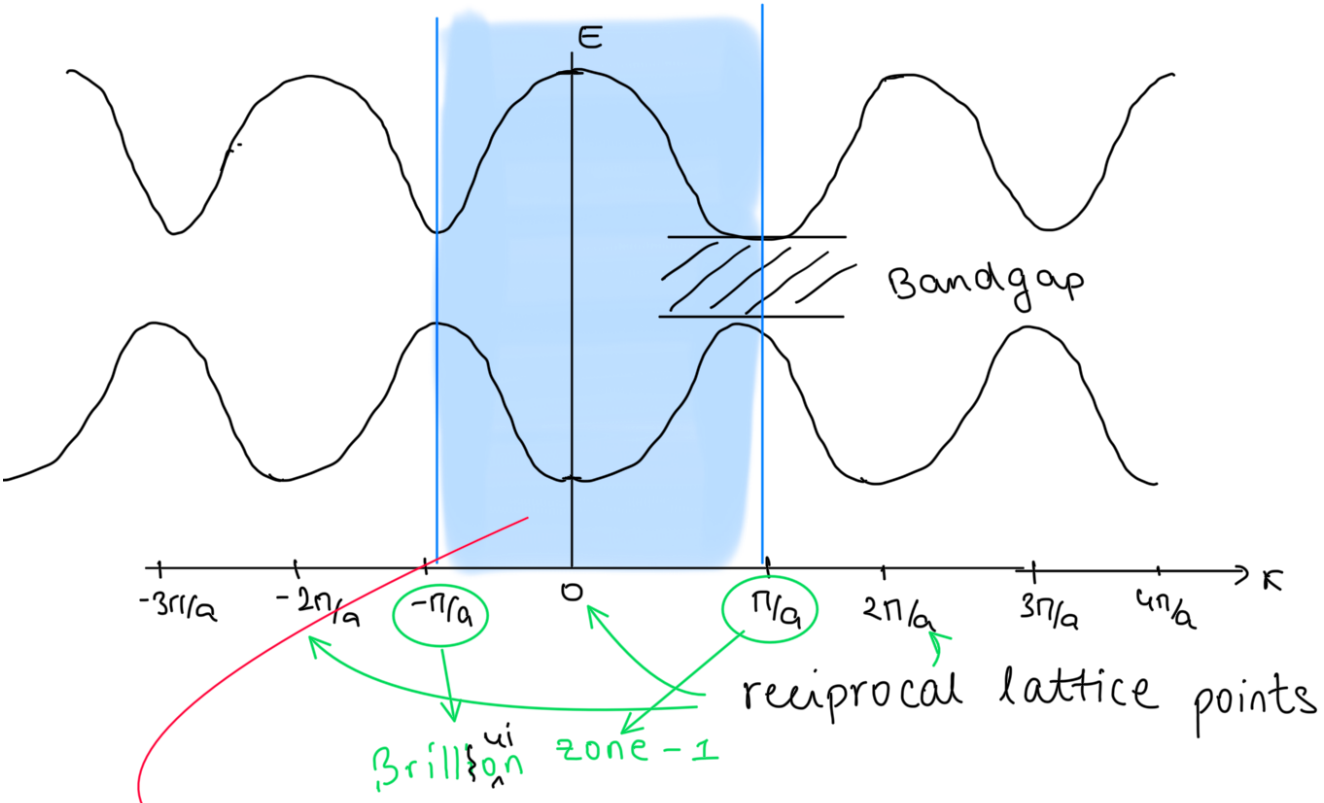
$k \rightarrow$ almost continuous $\qquad k = \frac{2\pi}{a}$





wave vector (k)

They aren't sinusoidal in general, they're periodic though



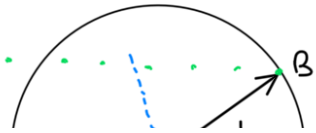
Reduced E-k diagram

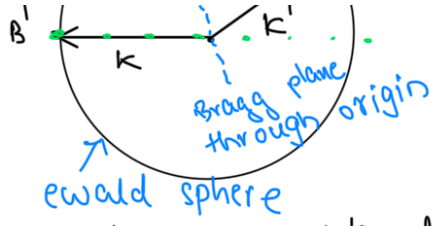
Diffraction in Brillouin zone boundary

Whenever a Bragg plane touches one end of wave vector, diffraction condition is satisfied. (whenever a wave vector touches BZ boundary)

k – wave vector

k't diffracted wave vector





for free $e^{\ominus}$, $E = \frac{\hbar^2 K^2}{2m}$. But after it enters a crystal, the energy gets quantized i.e there're forbidden gaps.

How bands are occupied by elections:

- @ 0K → top most completely filled band = valance band
- Band above valance band → Conduction band
- as TT, electrons gain energy and go to higher bands leaving behind vacancies

$$v_g \text{ (group velocity)} = \frac{d\omega}{dk} = \frac{1}{\hbar}\frac{dE}{dk} \quad (\because E = \hbar\omega)$$

$$\Rightarrow \frac{dE}{dk} = \hbar v_g$$

Electrons in conduction band

Holes in valence band

Small gap (electron moves from VB + CB)

semiconductor

Large gap → insulator

for metal, Fermi needs to be in conduction band

material	Bandgap (ev)
Ge	0.66

Si	1.12
GaAs	1.4
GaN	3.4
Ga_2O_3	4.9
Diamond	5.5
AlN	6.4

} ultrawide bandgap

E-k diagram — optics

E-x diagram — electron devices

direct bandgap semiconductors

indirect bandgap semiconductors

Effective mass: $F = m^* \frac{dv_g}{dt} = \hbar \frac{dk}{dt}$

$v_g = 1/\hbar \; dE/dk \quad dv_g/dt = 1/\hbar \; d/dk \left(dE/dk\right) dk/dt$

$\Rightarrow \frac{1}{m} = \frac{1}{\hbar^2} \frac{d^2E}{dk^2}$